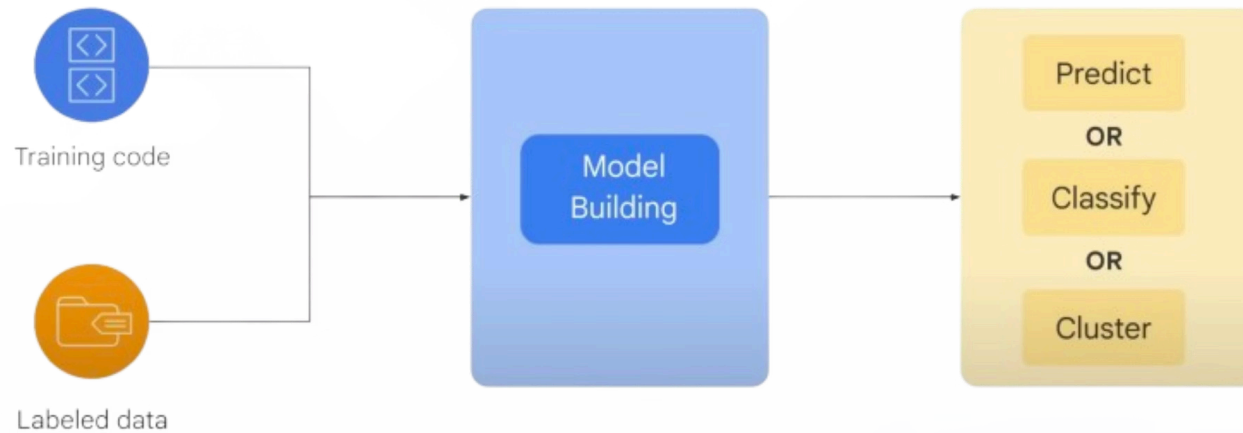




Testleaf

A l w a y s A h e a d



Supervised Learning

RECAP

In supervised learning, a model is trained on **labeled data** — pairs of inputs and known outputs. The training process builds an internal representation that can then be used to **predict**, **classify**, or **cluster** new, unseen inputs. This foundational paradigm underpins most real-world ML applications, from spam detection to medical diagnosis.

Unsupervised Learning

Unlike supervised learning, **unsupervised learning** works with **unlabeled data**. The model is not given explicit right or wrong answers — instead, it discovers hidden structure, patterns, and groupings on its own. This makes it powerful for exploratory analysis, anomaly detection, and recommendation engines where labeled data is scarce or expensive to obtain.

No Labels Required

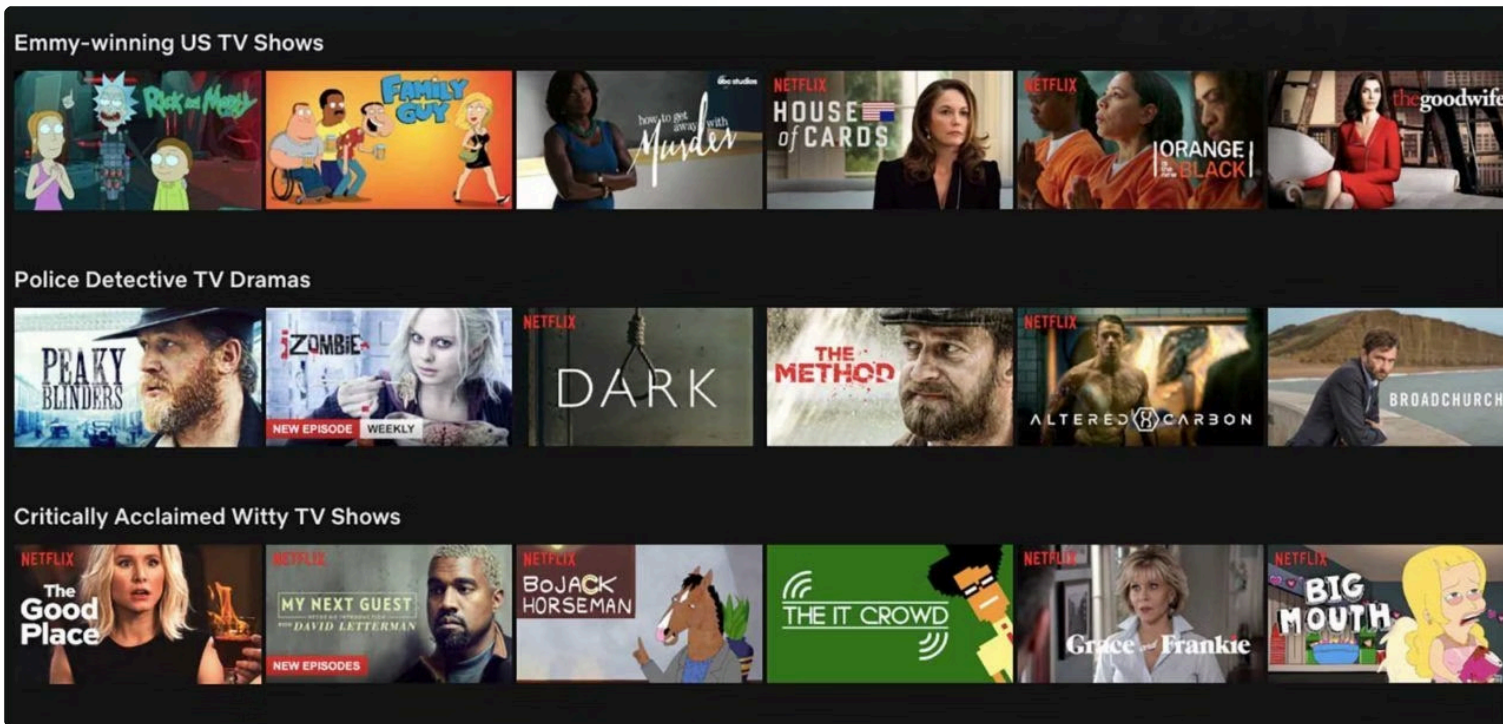
The algorithm finds structure without being told what to look for.

Pattern Discovery

Clusters, associations, and latent features emerge from raw data.

Real-World Scale

Works well when labeled datasets are too costly or unavailable.



Recommender Systems

UNLABELED PATTERN DISCOVERY

Recommender systems are one of the most visible applications of unsupervised learning. Platforms like **Netflix** and **Amazon** analyze vast amounts of user behavior data — without needing explicit labels — to surface content that a given user is likely to enjoy. The result: personalized suggestions that feel intuitive and drive engagement.

Collaborative Filtering

UNSUPERVISED LEARNING

How It Works

The model **groups users based on common behavior** — no labeled data required. Patterns emerge from similarity across users.

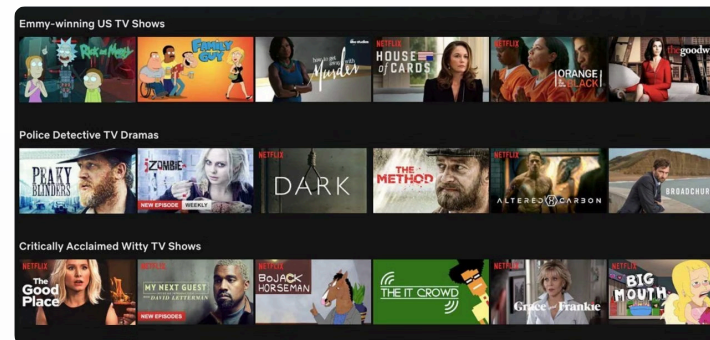
- No explicit labels needed
- Patterns discovered from similarity
- Scales to millions of users

Example in Action

If 100 users who watched "**Maharaja**" also watched "**Meiyazhagan**", the system automatically recommends "Meiyazhagan" to any new user who watched Maharaja — without anyone ever tagging the films as similar.



This is pure pattern discovery: the system infers relationships from behavior alone.



Content-Based Filtering

SUPERVISED LEARNING

How It Works

The model learns from **explicit user ratings or interactions**. Labels like star ratings, likes, and dislikes serve as ground truth during training.

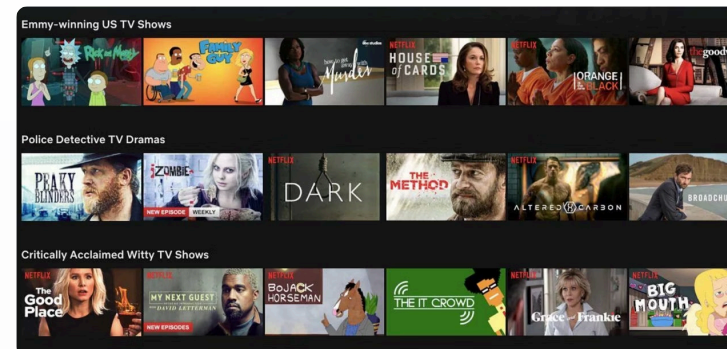
- Explicit labels exist (ratings, likes/dislikes)
- Structured training data available
- Learns user preference profiles

Example in Action

If a user gives a **5-star rating to "Maharaja"**, the system explicitly learns that this user prefers Thriller movies and surfaces similar content accordingly.



Explicit feedback makes the model's learning signal clearer and more targeted than collaborative approaches.



Netflix & Amazon Use Both

HYBRID APPROACH

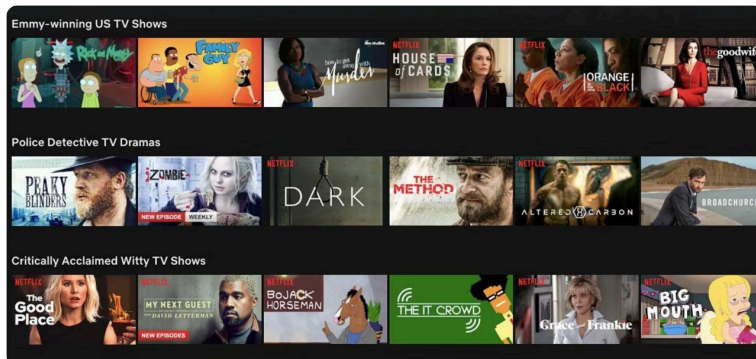
The most effective recommender systems don't choose one paradigm — they blend both. By combining unsupervised collaborative filtering with supervised content-based filtering, platforms can handle cold-start users *and* leverage rich explicit feedback at scale.

Unsupervised — Collaborative Filtering

Suggests items based on **user patterns and behavioral similarities**. No labels required. Works well at scale even for new content.

Supervised — Content-Based Filtering

Uses **explicit preferences** — ratings, purchase history, watch time — to fine-tune recommendations for individual users.





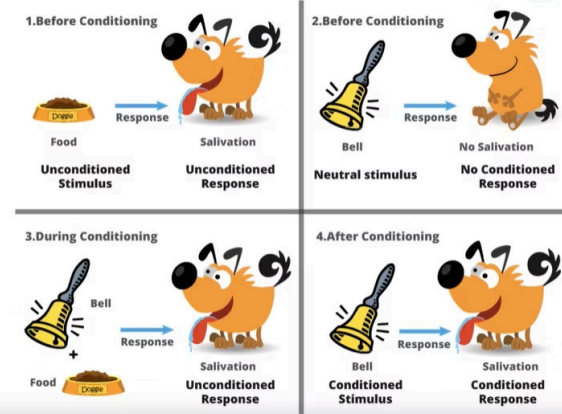
Reinforcement Learning

LEARNING BY REWARD LOOPS

In reinforcement learning, an **agent** learns by interacting with an environment and receiving **rewards or penalties** based on its actions. Unlike supervised learning, there are no labeled examples — the agent discovers optimal strategies through trial, error, and feedback. Self-driving cars, game-playing AI, and robotic control are prime examples of RL in the real world.

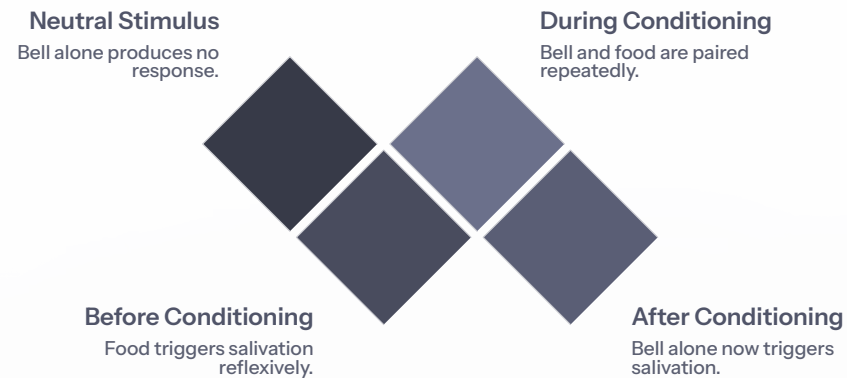
Reinforcement Learning

Conditioning



Reinforcement Learning — Conditioning

The classic **Pavlovian conditioning** analogy illustrates the RL loop clearly. Before conditioning, a neutral stimulus (a bell) produces no response. During conditioning, the bell is paired repeatedly with food. After conditioning, the bell alone triggers salivation — the agent has **learned a new association through repeated reward signals**. RL algorithms follow the same principle: repeated feedback shapes behavior over time.



AI vs. ML vs. DL

RECALL: THE FUNDAMENTALS

These three terms are often used interchangeably, but they describe a clear hierarchy. **Artificial Intelligence** is the broadest field — any technique that enables machines to mimic human cognition. **Machine Learning** is a subset of AI that learns from data. **Deep Learning** is a subset of ML using neural networks. Modern tools like ChatGPT, Llama, and Gemini sit at the intersection of Deep Learning, LLMs, and Generative AI.

